

Impact of Residual Transparency in Switchable-Opacity Displays for Video Passthrough Driving

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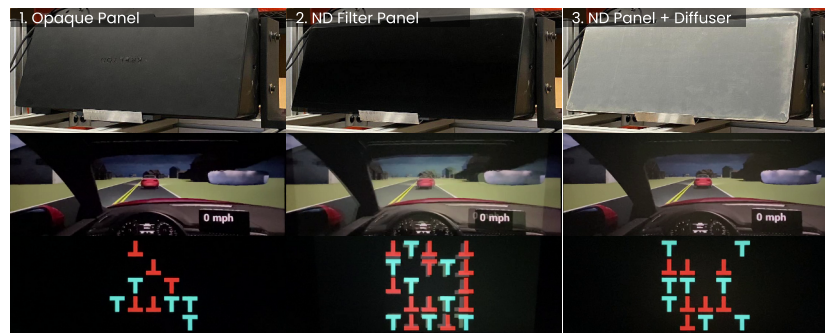


Figure 1: From left to right: (1) fully opaque back panel, (2) neutral-density (ND) filter back panel with residual transparency, and (3) ND filter back panel with an added diffuser. Top row shows the physical configurations mounted behind the display; middle row shows the driving task view; bottom row shows the visual search view. The ND filter panel causes a double image, the diffuser significantly reduces the artifact but still allows light bleed-through, and the opaque panel yields a single, artifact-free image.

ABSTRACT

Switchable-opacity displays promise for safer video passthrough (VP) by enabling emergency optical see-through; however, their opaque state may exhibit residual transparency, creating double images. We evaluate this phenomenon in a within-participant study ($N = 17$) that compares fully opaque, imperfectly opaque (ND filter), and ND-plus-diffuser conditions. Participants performed simulated driving and visual search tasks while subjective workload, eye strain, and eye-related physiological measures were recorded. Results show that residual transparency significantly increases perceived eye strain and workload, despite no degradation in task performance. The results also indicate these effects can be mitigated with a diffuser panel.

Index Terms: Residual Transparency, Double-Image Artifact, Switchable-Opacity Display, Video Passthrough Driving.

1 INTRODUCTION

Video passthrough (VP) is already used in vehicles through digital mirrors and backup cameras, offering advantages such as extended visibility and the ability to overlay assistive information. Yet VP remains relegated to infotainment screens, mainly because of the safety risk; if the VP system fails, the display will block the driver's direct view. A switchable-opacity display, opaque for VP and transparent in emergencies, could mitigate this issue. Transparent displays with liquid-crystal (LC) or electrochromic back panels enable such opacity switching, yet their "opaque" state typically allows residual background light to bleed through. This leakage produces double-image artifacts, and the effect is amplified in VP systems placed in the driver's field-of-view, because the displayed imagery is strongly correlated with the real scene [1].

This problem is not unique to VP driving. AR glasses with dimming modules (e.g. XREAL AR glasses) exhibit similar residual

transparency, and emerging AI glasses (e.g. Meta Ray-Ban Display) show live camera feeds over the real world in some applications. While VP driving motivates our study, opacity switching is also relevant to established AR/MR industrial and maintenance domains, where safety-critical tasks – such as operating heavy machinery – similarly require unobstructed direct vision. Despite the growing relevance, the impact of double-image artifacts in VP remains poorly investigated. The closest prior work examines optical ghost images in HUDs and focuses on basic legibility [2, 3].

In this study, we examine the effect of residual transparency on VP driving using a transparent display with three combinations of back panels, illustrated in Fig. 1. By measuring vehicle control alongside visual search performance, we assess driving-relevant behavior and subjective experience in the presence of double-image artifacts. We hope that these preliminary results can inform the design of future VP systems deployed in safety-critical contexts.

2 SWITCHABLE-OPACITY DISPLAY

We used a custom display based on a modified Brelyon Ultra Reality® Mini. The baseline system employs a birdbath configuration, consisting of an LC display, a beam splitter, and a concave mirror, to form a larger virtual image. In our variant, the concave mirror was replaced with a concave half-mirror, and the rear enclosure was substituted with an LC panel that can switch between transparent and opaque states. In this study, however, instead of using the LC panel, we changed the opacity by switching back panels for more granular control of optical characteristics.

3 USER STUDY

We conducted a within-participant study with 17 participants. Each participant experienced three display conditions in counterbalanced order: (1) perfect opacity using an opaque plastic panel, (2) imperfect opacity using a neutral-density (ND) filter panel, and (3) the ND filter combined with a diffuser panel (Fig. 1, top). This study received Institutional Review Board approval (WCG, ID 20251515).

The custom transparent display was positioned at seated eye height. Participants viewed the display while also seeing a projector screen located in front of them, which rendered identical imagery.

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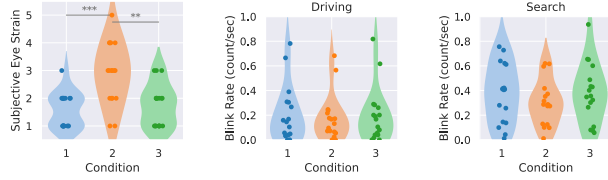


Figure 2: Comparison of eye strain: subjective rating (left), and blink rate during the driving task (middle) and the visual search task (right).

This configuration reproduced the double-image artifact caused by residual transparency in optical video passthrough driving. To control image overlap across participants, we instructed participants to adjust their chair height until a crosshair displayed at the center of both screens appeared spatially aligned.

Participants completed two tasks in each condition: a driving task (Fig. 1, middle) and a visual search task (Fig. 1, bottom). The driving task, implemented in Unreal Engine, presented a looped roadway with varying curvature and a lead vehicle moving at a constant speed. Participants were instructed to maintain a two-second headway to the lead vehicle using a Logitech G29 steering wheel and pedal set. The visual search task, adapted from Stoet's PsyToolkit [4], required participants to identify an orange upright "T" among distractors varying in color and orientation. Responses were made by pressing the space bar on a standard keyboard. During task execution, participants wore eye tracking glasses (Pupil Labs Neon) to record eye-related physiological signals. The whole study was conducted in a dark room without windows to maintain a consistent and controlled light environment.

Participants first completed a short practice drive using the opaque panel. For each experimental condition, they then performed the driving task (2 min) followed by the visual search task (~1.5 min). After each condition, participants completed the NASA-TLX and rated perceived eye strain using a subjective scale.

4 RESULTS

4.1 Eye Strain

A repeated-measures ANOVA revealed a large and significant effect of the transparency condition on subjective rating of eye strain, $F(2,32) = 19.25$, $p < 0.0086$, $\eta_G^2 = 0.324$. Post-hoc Holm-corrected comparisons showed that participants reported significantly higher eye strain for Condition 2 compared to Condition 1 ($p < 0.001$, $d = -1.44$) and Condition 3 ($p = 0.002$, $d = 1.18$), while the other two conditions did not differ, Fig. 2 (left). The same pattern was observed in the responses to "Eye strain negatively affected my ability to focus on these tasks" ($F(2,32) = 16.10$, $p < 0.001$, $\eta_G^2 = 0.216$), with Condition 2 reported higher impact than Conditions 1 and 3 (both $p < 0.001$).

A repeated-measures ANOVA indicated a main effect of condition on blink rate in the visual search task, $F(2,32) = 3.65$, $p = 0.037$, $\eta_G^2 = 0.027$, Fig. 2 (right). However, Holm-corrected post hoc comparisons did not reach significance, showing weak trends of lower blink rate for Condition 2 compared to Condition 1 ($p = 0.069$, $d = 0.34$) and Condition 3 ($p = 0.098$, $d = -0.37$). While lower blink rate can be interpreted in multiple ways, the subjective rating results indicate that this is due to higher visual demand rather than lower eye strain. There were no significant differences in driving task blink rate or in pupil diameter for either task.

4.2 Workload

Repeated-measures ANOVAs revealed a significant effect of condition for all NASA-TLX subscales (all $p \leq 0.019$), Fig. 3. Holm-corrected post-hoc tests showed that Condition 2 indicated higher

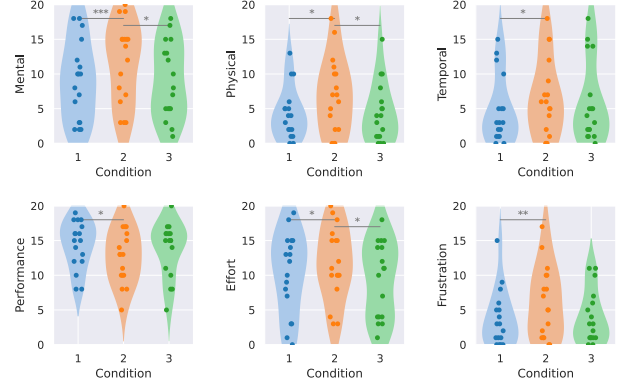


Figure 3: Comparison of Workload (NASA-TLX).

workload than Condition 1 across all subscales (Mental $p < 0.001$, Physical $p = 0.022$, Temporal $p = 0.034$, Performance $p = 0.011$, Effort $p = 0.029$, Frustration $p = 0.007$) with small to medium effect size ($0.25 \leq |d| \leq 0.57$). Compared to Condition 3, Condition 2 also showed higher Mental demand ($p = 0.012$, $d = 0.47$), Physical demand ($p = 0.018$, $d = 0.51$), and Effort ($p = 0.022$, $d = 0.42$), with no other contrasts reaching significance.

4.3 Task Performance

We conducted repeated-measures ANOVAs on vehicle control measures, including time headway, lateral offset, and pedal/steering smoothness, and none yielded significant effects of the transparency condition. Similarly, reaction time and accuracy in the visual search task did not show significant differences across conditions.

5 DISCUSSION

While all subjective measures indicated a negative impact of the double-image artifact in Condition 2, actual task performance metrics did not show notable differences. This suggests that, while participants generally needed to put in more effort, their performance in the tasks remained stable across conditions. This study further indicates that the use of a diffuser layer can mitigate this issue, effectively removing the presence of the background image.

One limitation of this study is the disparity in lighting conditions between the real outdoor environment and the laboratory darkroom. Since it is challenging to recreate daylight brightness, we tried to reproduce the background/foreground contrast instead by modifying the brightness of both screens. Additionally, future studies investigating longer periods of driving time, beyond the 2-minute driving task in our experiment, could provide a more realistic setting in which to study the effects of double-image-induced eyestrain. Despite the limitations, we believe that these initial results can provide guidance for designers of safety-critical VP systems.

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